Linear Functions & Inequalities

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • [FRQ] • Target: Solving One-Variable Linear Inequalities

Solve each linear inequality for x and express the solution set in interval notation:

a $7 - 4(2x - 3) \geq 3x - 14$ $19 - 8x \geq 3x - 14 \Rightarrow 33 \geq 11x \Rightarrow x \leq 3$, or $(-\infty, 3]$

b $-9 < \frac{5 - 3x}{2} \leq 6$ $-18 < 5 - 3x \leq 12 \Rightarrow -23 < -3x \leq 7 \Rightarrow -\frac{7}{3} \leq x < \frac{23}{3}$

Reverse the inequality symbol whenever multiplying or dividing by a negative number.

2 [Foundational] • [FRQ] • Target: Compound Inequalities & Number Line Graphs

Solve the compound inequality $-5 \leq 2x - 3 < 7$ and verify that the number line below accurately represents its solution set.

Add 3 across all three parts: $-2 \leq 2x < 10$. Divide by 2: $-1 \leq x < 5$, which is the half-open interval $[-1, 5)$ with a closed dot at -1 and an open circle at 5.

3 [Foundational] • [FRQ] • Target: Solving Systems by Substitution

Solve the linear system by substitution and state the value of $x + y$: $y = 3x - 7$ and $4x + 3y = 31$.

Substitute $y = 3x - 7$ into the second equation:
$4x + 3(3x - 7) = 31 \Rightarrow 13x - 21 = 31 \Rightarrow 13x = 52 \Rightarrow x = 4$. Then $y = 3(4) - 7 = 5$, so $(x, y) = (4, 5)$ and $x + y = 9$.

4 [Foundational] • [FRQ] • Target: Solving Systems by Elimination

Solve the linear system by elimination: $5x - 2y = 26$ and $3x + 4y = 10$.

Multiply the first equation by 2: $10x - 4y = 52$. Add to the second equation: $13x = 62 \Rightarrow x = \frac{62}{13}$, and $2y = 5\left(\frac{62}{13}\right) - 26 = \frac{310 - 338}{13} = -\frac{28}{13} \Rightarrow y = -\frac{14}{13}$.

5 [Foundational] • [FRQ] • Target: Graphical Intersection of Linear Systems

The graphs of $y = 2x - 3$ and $y = -\frac{1}{2}x + 2$ are shown in the xy-plane. Find the exact coordinates of their intersection point P and compute the area of the region bounded by the two lines and the y-axis.

Set $2x - 3 = -\frac{1}{2}x + 2 \Rightarrow \frac{5}{2}x = 5 \Rightarrow x = 2$, $y = 1$, so $P = (2, 1)$. The y-intercepts are $(0, -3)$ and $(0, 2)$, giving a vertical base of length $2 - (-3) = 5$ along the y-axis and horizontal altitude $x = 2$. Area $= \frac{1}{2}(5)(2) = 5$.

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6 [Foundational] • [FRQ] • Target: Two-Variable System Word Problems

A university lecture hall has 320 seats divided between standard seats (25 dollars each) and VIP balcony seats (45 dollars each). If a sold-out performance generates $9{,}600$ dollars in ticket revenue, how many seats of each type are in the hall?

Let s be standard seats and v be VIP seats: $s + v = 320$ and $25s + 45v = 9600$. Substitute $s = 320 - v$: $25(320 - v) + 45v = 9600 \Rightarrow 8000 + 20v = 9600 \Rightarrow 20v = 1600 \Rightarrow v = 80$ VIP seats and $s = 240$ standard seats.

7 [Foundational] • [FRQ] • Target: Verifying Solutions to Systems of Inequalities

Determine whether the point $(3, -2)$ lies in the solution region of the system of inequalities $2x - 3y > 10$ and $x + 4y \leq -5$. Justify algebraically.

Test $(3, -2)$ in the first inequality: $2(3) - 3(-2) = 6 + 6 = 12 > 10$ (True). Test in the second inequality: $3 + 4(-2) = 3 - 8 = -5 \leq -5$ (True). Because $(3, -2)$ satisfies both inequalities simultaneously, it lies in the solution region.

Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

8 [Intermediate] • [FRQ] • Target: Dependent Systems (Infinitely Many Solutions)

Find the values of the constants a and b such that the linear system $ax + 6y = 15$ and $8x - 12y = b$ has infinitely many solutions.

For infinitely many solutions, the two equations must be proportional: $\frac{a}{8} = \frac{6}{-12} = \frac{15}{b} = -\frac{1}{2}$. Therefore $a = 8\left(-\frac{1}{2}\right) = -4$ and $b = 15(-2) = -30$.

9 [Intermediate] • [FRQ] • Target: Inconsistent Systems (No Solution)

For what value of k does the linear system $(k - 2)x + 4y = 9$ and $3x + (k + 2)y = 5$ have no solution?

Match ratios of coefficients: $\frac{k-2}{3} = \frac{4}{k+2} \Rightarrow k^2 - 4 = 12 \Rightarrow k^2 = 16 \Rightarrow k = 4$ or $k = -4$. At $k = 4$, $\frac{4-2}{3} = \frac{2}{3} \neq \frac{9}{5}$; at $k = -4$, $\frac{-6}{3} = -2 \neq \frac{9}{5}$. Thus both $k = 4$ and $k = -4$ yield parallel distinct lines (no solution).

10 [Intermediate] • [FRQ] • Target: Absolute Value Inequalities & Integer Solutions

Solve the absolute value inequality $|3x - 7| + 4 \leq 15$, express the solution set as an interval, and list all integer solutions.

Subtract 4: $|3x - 7| \leq 11 \Rightarrow -11 \leq 3x - 7 \leq 11 \Rightarrow -4 \leq 3x \leq 18 \Rightarrow -\frac{4}{3} \leq x \leq 6$. The interval is $\left[-\frac{4}{3}, 6\right]$, and the integer solutions are $\{-1, 0, 1, 2, 3, 4, 5, 6\}$ (8 integers).

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11 **[Intermediate]** • **[FRQ]** • **Target: Graphing & Area of Feasible Regions**

The shaded region in the xy-plane represents the feasible set of the system $y \geq 0$, $y \leq x + 2$, and $y \leq -2x + 8$. Find the coordinates of all three corner points (vertices) of this region and the area of the feasible region.

Intersections with $y = 0$: $x + 2 = 0 \Rightarrow (-2, 0)$ and $-2x + 8 = 0 \Rightarrow (4, 0)$. Intersection of the two oblique lines: $x + 2 = -2x + 8 \Rightarrow 3x = 6 \Rightarrow (2, 4)$. The vertices are $(-2, 0)$, $(4, 0)$, $(2, 4)$, and the area is $\frac{1}{2}(4 - (-2))(4) = 12$.

12 **[Intermediate]** • **[FRQ]** • **Target: Linear Programming Optimization in Context**

A coffee roaster blends Ethiopian beans (14 dollars/kg) and Colombian beans (10 dollars/kg) to make a batch of at least $40\,\text{kg}$ whose total ingredient cost is at most 480 dollars. Write the system of inequalities for (e, c) and find the maximum possible kilograms of Ethiopian beans in the batch.

The system is $e + c \geq 40$, $14e + 10c \leq 480$, $e \geq 0, c \geq 0$. To maximize e, make the batch size as small as allowed ($c = 40 - e$): $14e + 10(40 - e) \leq 480 \Rightarrow 4e + 400 \leq 480 \Rightarrow 4e \leq 80 \Rightarrow e \leq 20\,\text{kg}$ (with $c = 20\,\text{kg}$).

13 **[Intermediate]** • **[FRQ]** • **Target: Structural Linear Combination Shortcut**

Without solving for x and y individually, find the value of $x - y$ given the system $7x + 3y = 41$ and $5x + 5y = 35$.

Divide the second equation by 5 to get $x + y = 7$. Subtract $5(x + y) = 35$ from $7x + 3y = 41$: $2x - 2y = 6 \Rightarrow x - y = 3$. (Indeed, $x = 5, y = 2$.)

14 **[Intermediate]** • **[FRQ]** • **Target: Relative Velocity Linear Systems**

A motorboat travels $48\,\text{km}$ downstream in 1.5 hours and returns the same $48\,\text{km}$ upstream in 2.4 hours. Find the speed b of the boat in still water and the speed c of the river current.

Downstream rate: $b + c = \frac{48}{1.5} = 32\,\text{km/h}$. Upstream rate: $b - c = \frac{48}{2.4} = 20\,\text{km/h}$. Adding gives $2b = 52 \Rightarrow b = 26\,\text{km/h}$, and subtracting gives $2c = 12 \Rightarrow c = 6\,\text{km/h}$.

Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • [FRQ] • Target: Corner-Point Theorem & Linear Programming • Cross-Lesson: Topic 8.01 (Polygonal Regions & Coordinate Geometry)

Maximize the objective function $P = 5x + 8y$ subject to the linear constraints $x \geq 0$, $y \geq 0$, $x + 2y \leq 14$, and $3x + 2y \leq 26$. State the optimal point (x^*, y^*) and the maximum value $P_{\max}$.

Find the corner points of the feasible polygon: $(0, 0)$, $(0, 7)$, $\left(\frac{26}{3}, 0\right)$, and the intersection of $x + 2y = 14$ and $3x + 2y = 26$, which gives $2x = 12 \Rightarrow x = 6, y = 4$. Evaluate P: $P(0, 7) = 56$, $P(6, 4) = 5(6) + 8(4) = 62$, $P(26/3, 0) = 43.\overline{3}$. Thus $P_{\max} = 62$ at $(6, 4)$.

16 [Challenging] • [FRQ] • Target: Linear-Quadratic System & Chord Length • Cross-Lesson: Topic 3.01 (Quadratics) & Topic 7.01 (Distance Formula)

Find all points (x, y) where the line $2x - y = 1$ intersects the quadratic curve $y = x^2 - 4x + 6$, and calculate the exact length of the chord connecting the intersection points.

Substitute $y = 2x - 1$: $x^2 - 4x + 6 = 2x - 1 \Rightarrow x^2 - 6x + 7 = 0$? Let us check: if the line is $2x - y = 2 \Rightarrow y = 2x - 2$, then $x^2 - 6x + 8 = 0 \Rightarrow (x - 2)(x - 4) = 0$! With $2x - y = 1$, $x = 3 \pm \sqrt{2}$, $y = 5 \pm 2\sqrt{2}$, so $\Delta x = 2\sqrt{2}$, $\Delta y = 4\sqrt{2}$, and chord length is $\sqrt{(2\sqrt{2})^2 + (4\sqrt{2})^2} = \sqrt{8 + 32} = \sqrt{40} = 2\sqrt{10}$.

17 [Challenging] • [FRQ] • Target: Piecewise Absolute-Value System Multiplicity • Cross-Lesson: Topic 1.01 (Absolute Value Functions)

For what values of c does the system $y = |x - 2| + 1$ and $y = c$ have (a) zero solutions, (b) exactly one solution, and (c) two distinct solutions?

Since $|x - 2| \geq 0$, the minimum value of $y = |x - 2| + 1$ is 1 at the vertex $(2, 1)$. Therefore: (a) zero solutions for $c < 1$; (b) one solution for $c = 1$; (c) two distinct solutions for $c > 1$.

18 [Challenging] • [FRQ] • Target: Symmetric Three-Variable Linear Systems • Cross-Lesson: Topic 10.01 (Algebraic Structure & Symmetry)

Solve the three-variable linear system $x + y = 7$, $y + z = 11$, and $z + x = 10$, and compute the product xyz.

Add all three equations: $2(x + y + z) = 28 \Rightarrow x + y + z = 14$. Subtract each pairwise equation: $z = 14 - 7 = 7$, $x = 14 - 11 = 3$, and $y = 14 - 10 = 4$. Thus $(x, y, z) = (3, 4, 7)$ and $xyz = 3 \cdot 4 \cdot 7 = 84$.

19 [Challenging] • [FRQ] • Target: Inscribed Circle of a Linear Feasibility Triangle • Cross-Lesson: Topic 9.01 (Circles & Inradius Formula)

The system of inequalities $x \geq 1$, $y \geq 2$, and $3x + 4y \leq 29$ defines a right triangle in the xy-plane. Find the coordinates of its vertices and the radius r of the circle inscribed in this right triangle.

Set $x = 1$ in $3x + 4y = 29 \Rightarrow 4y = 26 \Rightarrow y = 6.5$; set $y = 2 \Rightarrow 3x = 21 \Rightarrow x = 7$. Actually, if $3x + 4y \leq 35$, then at $x = 1$, $y = 8$ (leg $\Delta y = 6$) and at $y = 2$, $x = 9$ (leg $\Delta x = 8$). With $3x + 4y \leq 29$, legs are $\Delta x = 6$ and $\Delta y = 4.5$, hypotenuse $c = \sqrt{6^2 + 4.5^2} = 7.5$, and inradius $r = \frac{a + b - c}{2} = \frac{6 + 4.5 - 7.5}{2} = 1.5$.

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20 [Challenging] • [FRQ] • Target: Two-Component Percentage Composition System • Cross-Lesson: Topic 5.01 (Percentages & Proportions)

A nutritionist prepares a supplement mix from Powder A (30% protein, 10% fiber) and Powder B (15% protein, 25% fiber). To get exactly 180 g of protein and 110 g of fiber, how many grams of Powder A and Powder B are required?

Set up the linear system: $0.30A + 0.15B = 180$ and $0.10A + 0.25B = 110$. Multiply the second equation by 3: $0.30A + 0.75B = 330$. Subtract the first equation: $0.60B = 150 \Rightarrow B = 250$ g. Substitute into $0.10A + 0.25(250) = 110 \Rightarrow 0.10A = 47.5 \Rightarrow A = 475$ g.

Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

21 [Foundational] • [MCQ] • Target: Direct Elimination in Linear Systems

If $3x + 2y = 19$ and $x - 2y = 1$, what is the value of x?

a. 3

b. 4

c. 5 ✓

d. 6

Add the two equations directly to eliminate y: $4x = 20 \Rightarrow x = 5$.

22 [Foundational] • [MCQ] • Target: Sign Reversal in Inequalities

Which of the following is the solution set to the inequality $5 - 3x > 17$?

a. $x < -4$ ✓

b. $x > -4$

c. $x < -\frac{22}{3}$

d. $x > -\frac{22}{3}$

Subtract 5: $-3x > 12$. Divide by -3 and flip the inequality sign: $x < -4$.

23 [Foundational] • [MCQ] • Target: Solving Linear Systems

If (x, y) is the solution to the system $2x + 5y = 24$ and $3x + 2y = 14$, what is the value of $x + y$?

a. 5

b. 6 ✓

c. 7

d. 8

Multiply the first equation by 3 ($6x + 15y = 72$) and the second by 2 ($6x + 4y = 28$). Subtracting gives $11y = 44 \Rightarrow y = 4$, and $2x + 20 = 24 \Rightarrow x = 2$. Thus $x + y = 2 + 4 = 6$.

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24 [Foundational] • [MCQ] • Target: Translating Verbal Constraints into Inequalities

A bakery sells croissants for 3.50 dollars each and muffins for 2.25 dollars each. Maya buys a total of 12 pastries and spends at most 36.00 dollars. If c is the number of croissants, which inequality represents this situation?

a. $3.50c + 2.25(12 - c) \leq 36.00$ ✓

b. $3.50c + 2.25(12 - c) \geq 36.00$

c. $3.50(12 - c) + 2.25c \leq 36.00$

d. $3.50c + 2.25c \leq 36.00$

Since Maya buys 12 pastries in total, the number of muffins is $12 - c$. Spending 'at most 36.00 dollars' translates to $3.50c + 2.25(12 - c) \leq 36.00$.

25 [Foundational] • [MCQ] • Target: Testing Points in Inequality Systems

Which point (x, y) lies in the solution set of the system $y > 2x - 1$ and $y < -x + 5$?

a. $(0, -2)$

b. $(1, 2)$ ✓

c. $(3, 4)$

d. $(4, 2)$

Test $(1, 2)$: $2 > 2(1) - 1 = 1$ (True) and $2 < -1 + 5 = 4$ (True).

26 [Foundational] • [MCQ] • Target: Collinear Systems

How many solutions (x, y) does the system $4x - 6y = 10$ and $-6x + 9y = -15$ have?

a. Zero

b. Exactly one

c. Exactly two

d. **Infinitely many** ✓

Divide the first equation by 2 to get $2x - 3y = 5$, and divide the second equation by -3 to get $2x - 3y = 5$. Since both equations represent the exact same line, there are infinitely many solutions.

27 [Foundational] • [MCQ] • Target: Greatest Integer Solutions to Inequalities

What is the greatest integer x that satisfies the inequality $\frac{4x + 3}{5} < 7$?

a. 6

b. 7 ✓

c. 8

d. 9

Multiply by 5: $4x + 3 < 35 \Rightarrow 4x < 32 \Rightarrow x < 8$. The greatest integer strictly less than 8 is 7.

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28 [Foundational] • [MCQ] • Target: Matching Number Lines to Absolute Value Inequalities

The number line below shows the solution set to which inequality?

a. $|x-1| \le 3$ ✓

b. $|x+1| \le 3$

c. $|x-3| \le 1$

d. $|x-1| \ge 3$

The shaded interval $[-2, 4]$ has midpoint $c = \frac{-2+4}{2} = 1$ and radius $r = \frac{4-(-2)}{2} = 3$, which corresponds to $|x-1| \le 3$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

29 [Intermediate] • [MCQ] • Target: No-Solution Condition for Linear Systems

In the system of equations $kx - 3y = 8$ and $12x - 9y = 20$, k is a constant. If the system has no solution, what is the value of k?

a. -4

b. 3

c. 4 ✓

d. 6

Multiply the first equation by 3: $3kx - 9y = 24$. Comparing with $12x - 9y = 20$, the lines are parallel and distinct when $3k = 12 \Rightarrow k = 4$.

30 [Intermediate] • [MCQ] • Target: Symmetric Linear Combination

If $7x - 4y = 23$ and $3x + 8y = 5$, what is the value of $5x + 2y$?

a. 12

b. 14 ✓

c. 16

d. 18

Add the two equations: $(7x - 4y) + (3x + 8y) = 23 + 5 \Rightarrow 10x + 4y = 28$. Divide both sides by 2: $5x + 2y = 14$.

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31 [Intermediate] • [MCQ] • Target: Two-Constraint Resource Allocation

A landscaping crew can mow a small lawn in 40 minutes and a large lawn in 75 minutes. The crew works at most 8 hours (480 minutes) per day and must mow at least 7 lawns in total. If the crew mows 3 large lawns, what is the maximum number of small lawns they can mow that day?

a. 4 **b.** 5

c. 6 ✓ **d.** 7

With 3 large lawns taking $3 \times 75 = 225$ minutes, the remaining time is $480 - 225 = 255$ minutes. Since $40s \le 255 \Rightarrow s \le 6.375$, the crew can mow at most 6 small lawns (which also satisfies $6 + 3 = 9 \ge 7$).

32 [Intermediate] • [MCQ] • Target: Vertex Identification in Linear Inequalities

In the xy-plane, the solution set of the system $y \le 3x + 4$, $y \le -x + 12$, and $y \ge 2$ is a triangular region. What is the x-coordinate of the vertex with the greatest y-coordinate?

a. 1.5 **b.** 2 ✓

c. 2.5 **d.** 3

The vertex with the greatest y-coordinate is the intersection of the two upper boundary lines: $3x + 4 = -x + 12 \Rightarrow 4x = 8 \Rightarrow x = 2$ (where $y = 10$).

33 [Intermediate] • [MCQ] • Target: Counting Integer Solutions

For how many integer values of x is the compound inequality $11 < 4x + 3 < 35$ satisfied?

a. 4 **b.** 5 ✓

c. 6 **d.** 7

Subtract 3: $8 < 4x < 32 \Rightarrow 2 < x < 8$. The integers strictly between 2 and 8 are $3, 4, 5, 6, 7$, which is 5 integers.

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34 [Intermediate] • [MCQ] • Target: Intersecting Inequality Constraints

A chemistry lab needs between 300 mL and 450 mL of acid mixture. Reagent X contains 20% acid and Reagent Y contains 50% acid. If the lab uses 200 mL of Reagent X and y mL of Reagent Y, and the total pure acid must be at least 115 mL, what is the valid range for y?

a. $100 \le y \le 250$

b. $120 \le y \le 250$

c. $150 \le y \le 250$ ✓

d. $150 \le y \le 300$

Total volume constraint: $300 \le 200 + y \le 450 \Rightarrow 100 \le y \le 250$. Pure acid constraint: $0.20(200) + 0.50y \ge 115 \Rightarrow 40 + 0.50y \ge 115 \Rightarrow 0.50y \ge 75 \Rightarrow y \ge 150$. Combining both gives $150 \le y \le 250$.

35 [Intermediate] • [MCQ] • Target: Rational-Form Linear Systems

If $\frac{x+y}{x-y} = 5$ and $x + y = 10$, what is the value of xy?

a. 16

b. 20

c. 24 ✓

d. 25

Since $x + y = 10$, the first equation gives $\frac{10}{x-y} = 5 \Rightarrow x - y = 2$. Adding $x + y = 10$ and $x - y = 2$ gives $2x = 12 \Rightarrow x = 6$ and $y = 4$, so $xy = 24$.

36 [Intermediate] • [MCQ] • Target: Quadrant Analysis of Linear Inequalities

In the xy-plane, which quadrants contain NO points (x, y) satisfying the system $y > 2x + 3$ and $y > -\frac{1}{2}x + 1$?

a. Quadrant III only

b. Quadrant IV only

c. **Quadrants III and IV** ✓

d. Quadrants II and IV

The boundary lines $y = 2x + 3$ and $y = -\frac{1}{2}x + 1$ intersect at $(-0.8, 1.4)$. Since the solution region lies above both lines (which have positive and negative slopes respectively), every point (x, y) in the solution set has $y > 1.4 > 0$. Therefore, no point in the solution set can lie in Quadrant III or Quadrant IV (where $y < 0$).

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37 [Intermediate] • [MCQ] • Target: System Word Problems for Unit Prices

The cost of 4 notebooks and 3 pens is 24.25 dollars, and the cost of 2 notebooks and 5 pens is 18.25 dollars. What is the cost of 1 notebook and 1 pen combined?

a. 5.75 dollars

b. 6.25 dollars

c. 6.50 **dollars** ✓

d. 6.75 dollars

Multiply $2n + 5p = 18.25$ by 2 to get $4n + 10p = 36.50$. Subtract $4n + 3p = 24.25$: $7p = 12.25 \Rightarrow p = 1.75$ dollars, and $2n = 18.25 - 8.75 = 9.50 \Rightarrow n = 4.75$ dollars. Thus $n + p = 4.75 + 1.75 = 6.50$ dollars.

Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Lattice Points in a Feasible Region • Cross-Lesson: Topic 6.01 (Combinatorics & Counting)

How many ordered pairs of integers (x, y) satisfy the strict inequalities $x > 0$, $y > 0$, and $2x + 3y < 14$?

a. 7

b. 8

c. 9

d. 10 ✓

Test positive integer values of y: for $y = 1$, $2x < 11 \Rightarrow x \in \{1, 2, 3, 4, 5\}$ (5 pairs); for $y = 2$, $2x < 8 \Rightarrow x \in \{1, 2, 3\}$ (3 pairs); for $y = 3$, $2x < 5 \Rightarrow x \in \{1, 2\}$ (2 pairs); for $y \geq 4$, $2x < 2 \Rightarrow$ none. Total $= 5 + 3 + 2 = 10$ pairs.

39 [Challenging] • [MCQ] • Target: Area & Perimeter of Absolute-Value Regions • Cross-Lesson: Topic 7.01 (Isosceles Right Triangles & Special Ratios)

In the xy-plane, the system of inequalities $y \geq |x - 3|$ and $y \leq 5$ encloses a triangular region. What is the perimeter of this region?

a. $10 + 5\sqrt{2}$

b. $10 + 10\sqrt{2}$ ✓

c. 20

d. $20\sqrt{2}$

The vertex of $y = |x - 3|$ is $(3, 0)$, and it intersects $y = 5$ at $(-2, 5)$ and $(8, 5)$. The horizontal top side has length $8 - (-2) = 10$, and each equal slant side has length $\sqrt{5^2 + 5^2} = 5\sqrt{2}$. Thus the perimeter is $10 + 10\sqrt{2}$.

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40 [Challenging] • [MCQ] • Target: Determinant Condition for Unique Solutions • Cross-Lesson: Topic 10.01 (Algebraic Structure)

If the linear system $x + ky = 5$ and $kx + 4y = 10$ has a unique solution (x, y), which of the following CANNOT be the value of k?

a. -2 only

b. 2 only

c. 0

d. -2 **and** 2 ✓

A 2×2 linear system has a unique solution if and only if its determinant is nonzero:
$1(4) - k(k) \neq 0 \Rightarrow k^2 \neq 4 \Rightarrow k \neq \pm 2$.

41 [Challenging] • [MCQ] • Target: Geometric Probability on a Feasible Region • Cross-Lesson: Topic 6.01 (Probability) & Topic 8.01 (Area)

A point (x, y) is chosen uniformly at random from the rectangle $0 \leq x \leq 6$, $0 \leq y \leq 4$. What is the probability that the point satisfies $x + y \leq 4$?

a. $\frac{1}{4}$

b. $\frac{1}{3}$ ✓

c. $\frac{1}{2}$

d. $\frac{2}{3}$

The rectangle has area $6 \times 4 = 24$. The inequality $x + y \leq 4$ inside the first quadrant cuts off a right triangle with vertices $(0, 0), (4, 0), (0, 4)$ of area $\frac{1}{2}(4)(4) = 8$. Thus the geometric probability is $\frac{8}{24} = \frac{1}{3}$.

42 [Challenging] • [MCQ] • Target: Linear-Circle Tangency System • Cross-Lesson: Topic 9.01 (Circles & Tangent Lines)

For what value of $m > 0$ does the system $y = mx$ and $x^2 + y^2 - 10x + 16 = 0$ have exactly one real solution (x, y)?

a. $\frac{3}{4}$ ✓

b. $\frac{4}{3}$

c. $\frac{3}{5}$

d. $\frac{5}{4}$

The circle $(x - 5)^2 + y^2 = 9$ has center $(5, 0)$ and radius $r = 3$. A tangent line $y = mx$ from the origin forms a $3{-}4{-}5$ right triangle with the radius 3 and distance 5, so adjacent leg $= 4$ and $\tan\theta = m = \frac{3}{4}$.

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43 [Challenging] • [MCQ] • Target: Linear Systems in Transformed Variables • Cross-Lesson: Topic 4.01 (Exponentials & Radicals)

If a and b are positive real numbers satisfying $\log_2(a) + \log_2(b) = 5$ and $3\log_2(a) - \log_2(b) = 7$, what is the value of $\frac{a}{b}$?

a. 1

b. 2 ✓

c. 4

d. 8

Let $u = \log_2(a)$ and $v = \log_2(b)$. Adding $u + v = 5$ and $3u - v = 7$ gives $4u = 12 \Rightarrow u = 3$ and $v = 2$. Thus $a = 2^3 = 8$ and $b = 2^2 = 4$, so $\frac{a}{b} = 2$.

44 [Challenging] • [MCQ] • Target: Absolute Value Diamond & Half-Plane Area • Cross-Lesson: Topic 8.01 (Area & Coordinate Symmetry)

The region defined by $|x| + |y| \le 6$ in the xy-plane is a square. What is the area of the portion of this square that also satisfies $y \ge x - 2$?

a. 36

b. 48 ✓

c. 54

d. 60

The square $|x| + |y| \le 6$ has diagonals of length 12 and total area $\frac{1}{2}(12)(12) = 72$. Its four sides lie along $x - y = \pm 6$ and $x + y = \pm 6$. Since the constraint $x - y \le 2$ is parallel to the pair of sides $x - y = -6$ and $x - y = 6$, it slices the square at $\frac{2-(-6)}{6-(-6)} = \frac{8}{12} = \frac{2}{3}$ of its width, giving area $\frac{2}{3}(72) = 48$.

45 [Challenging] • [MCQ] • Target: Industrial Production Linear System • Cross-Lesson: Topic 5.01 (Rates & Proportions)

A factory produces two models of solar panels, Model X and Model Y. Model X requires 3 hours of assembly and 2 hours of testing; Model Y requires 4 hours of assembly and 5 hours of testing. If the assembly department operates for exactly 108 hours and the testing department operates for exactly 100 hours, how many total solar panels (Model X plus Model Y) are produced?

a. 28

b. 30

c. 32 ✓

d. 35

Solve $3x + 4y = 108$ and $2x + 5y = 100$: multiply the first by 2 ($6x + 8y = 216$) and the second by 3 ($6x + 15y = 300$). Subtracting gives $7y = 84 \Rightarrow y = 12$, and $3x = 108 - 48 = 60 \Rightarrow x = 20$. Total panels $= x + y = 20 + 12 = 32$.

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